

INSA de Toulouse
Département GMM
BE - Processus de Poisson et Application en actuariat et fiabilité - 4 MA
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The Cramér-Lundberg modelling of risk in non-life insurance

Keywords : insurance, thinning method

Context of the project

The main goal of this project is to study a modeling of wealth/risk in actuarial sciences called the Cramér-Lundberg model. More precisely, the wealth of an insurance company can be modeled in a simple way as the difference of inwards (represented by the parameters $u, p \geq 0$) and outwards cashflows involving a compound Poisson process of the following form

$$R_t = u + pt - \sum_{i=1}^{N_t} Y_i, \quad t \geq 0; \quad (1)$$

where N is a Poisson process with constant intensity $\mu > 0$ and where the non-negative iid random variables $(Y_i)_{i \geq 1}$ represent the values of the sinistres involved in the contract. The goal of this project is to investigate what sort of risk analysis can be carried out in this model. More precisely, assuming a given common law on the variables Y_i 's, how one can choose u, p so that the value of R is controlled in some way ? Or put it differently, what criterion one could think of for measuring the risk of R ?

The main purpose of this project is thus to study some theoretical aspects of this model together to address the previous questions together with a numerical analysis of it. The numerical simulations will be presented in a Notebook Python.

Work to be performed

The work asked to the students goes in two different directions. Appropriate ressources will be provided to the students.

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Theoretical results

It is expected to study the different notions of risk management for giving a meaning of price in this model. This will cover the obtaining of the so-called safety loading coefficient and the study of the notion of ruin probability. If time allows, an introduction to large deviation principle and its application to the ruin theory will be done.

Numerical results

Numerical simulations presented in a Python Notebook are expected to illustrate the obtained theoretical results.

References

- [1] S. Asmussen and H. Albrecher. *Ruin Probabilities*. WORLD SCIENTIFIC, 2nd edition, 2010.